Experiment No. 9

Experiment Title	To implement Association Rule Mining (Apriori Algorithm) using Python
Course Outcomes	CO4: Implement the appropriate data mining methods like classification, clustering or Frequent Pattern mining on large data sets. CO5: Define and apply metrics to measure the performance of various data mining algorithms.
Date of Preparation	
Date of Submission	
Signature & Grade	

1. Aim

To implement the Apriori algorithm for Association Rule Mining in Python (mlxtend) on airline passenger data, and to find frequent combinations of passenger and service attributes and the rules between them, using support, confidence and lift.

2. Theory

2.1 Association Rule Mining

Association rule mining finds items that occur together in many transactions. A rule has the form IF antecedent THEN consequent, for example IF Class = Business AND Online boarding = High THEN satisfaction = satisfied. In this experiment a transaction is one passenger and an item is one attribute value of that passenger.

2.2 Apriori Algorithm

Apriori works in two steps. First it finds all frequent itemsets, which are combinations of items whose support is at least the minimum support. It starts with single items, keeps only the frequent ones, and builds larger combinations only from them. This is possible because of the Apriori property: every subset of a frequent itemset must also be frequent, so any combination that contains an infrequent item can be dropped. Second, it splits each frequent itemset into rules and keeps those whose confidence is at least the minimum confidence.

2.3 Measures

Measure	Formula	Meaning
Support	support(A, B) = transactions with A and B / all transactions	How common the rule is: the share of passengers that have all items of the rule.
Confidence	confidence(A → B) = support(A, B) / support(A)	Among passengers that have A, the share that also have B.
Lift	lift(A → B) = confidence(A → B) / support(B)	How much more often B occurs with A than by chance. Lift above 1 means a positive association.

3. Dataset and Tools

Dataset: Airline Passenger Satisfaction (Kaggle), file train.csv, with 103,904 passengers and 25 columns.

Source: kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

Apriori needs items, so 14 columns were turned into 39 items. Numeric columns were grouped into bands, and each value became an item written as Column=Value. A service rating of 0 means 'not applicable' in this dataset, so it creates no item.

Column(s)	Items created
Gender	Female, Male
Customer Type	Loyal, Disloyal
Type of Travel	Business, Personal

Column(s)	Items created
Class	Eco, Eco Plus, Business
Age	Under 25, 25-44, 45-64, 65+
Flight Distance	Short (up to 800), Medium (801 to 2,000), Long (over 2,000)
Online boarding, Inflight wifi service, Inflight entertainment, Seat comfort, Cleanliness, Leg room service	Low (rating 1-2), Medium (3), High (4-5), for each of the six columns
Departure Delay	None (0 min), Short (1-30 min), Long (over 30 min)
satisfaction	satisfied, neutral or dissatisfied

Tools: Python 3 with pandas, NumPy, Matplotlib, Seaborn and mlxtend.

4. Implementation

4.1 Load the Dataset

```
df = pd.read_csv('train.csv')
df.head()
```

	Unnamed: 0	id	Gender	Customer Type	Age	Type of Travel	Class	Flight Distance	Inflight wifi service	Departure/Arrival time convenient	Ease of Online booking	Gate location	Food and drink	Online boarding	Seat comfort	Inflight entertainment	On-board service	Le roc servic
0	0	70172	Male	Loyal Customer	13	Personal Travel	Eco Plus	460	3	4	3	1	5	3	5	5	4	
1	1	5047	Male	disloyal Customer	25	Business travel	Business	235	3	2	3	3	1	3	1	1	1	
2	2	110028	Female	Loyal Customer	26	Business travel	Business	1142	2	2	2	2	5	5	5	5	4	
3	3	24026	Female	Loyal Customer	25	Business travel	Business	562	2	5	5	5	2	2	2	2	2	
4	4	119299	Male	Loyal Customer	61	Business travel	Business	214	3	3	3	3	4	5	5	3	3	

Figure 1: The first rows of the original dataset

4.2 Create the Transactions

```
items_df['Age'] = pd.cut(df['Age'], bins=[0, 24, 44, 64, 120],
                         labels=['Under 25', '25-44', '45-64', '65+'])
for c in rating_cols:
    items_df[c] = pd.cut(df[c], bins=[0, 2, 3, 5], labels=['Low', 'Medium', 'High'])
basket = pd.get_dummies(items_df, prefix_sep='=', dtype=bool)
```

```
Transactions: 103904 | Items: 39 | Average items per transaction: 13.9
Columns turned into items: ['Gender', 'Customer Type', 'Type of Travel', 'Class', 'Age', 'Flight Distance', 'Online boarding', 'Inflight wifi service', 'Inflight entertainmen
```

	Gender	Customer Type	Type of Travel	Class	Age	Flight Distance	Online boarding	Inflight wifi service	Inflight entertainment	Seat comfort	Cleanliness	Leg room service	Departure Delay	satisfaction
0	Male	Loyal	Personal	Eco Plus	Under 25	Short	Medium	Medium	High	High	High	Medium	Short (1-30 min)	neutral or dissatisfied
1	Male	Disloyal	Business	Business	25-44	Short	Medium	Medium	Low	Low	Low	High	Short (1-30 min)	neutral or dissatisfied
2	Female	Loyal	Business	Business	25-44	Medium	High	Low	High	High	High	Medium	None	satisfied
3	Female	Loyal	Business	Business	25-44	Short	Low	Low	Low	Low	Low	High	Short (1-30 min)	neutral or dissatisfied
4	Male	Loyal	Business	Business	45-64	Short	High	Medium	Medium	High	Medium	High	None	satisfied

Figure 2: The transactions — number of items and the first rows after grouping

Each passenger is one transaction. The 14 columns became 39 items, and each transaction holds 13.9 items on average. The average is slightly below 14 because a rating of 0 creates no item.

4.3 Item Frequency

```
item_support = basket.mean().sort_values(ascending=False)
```

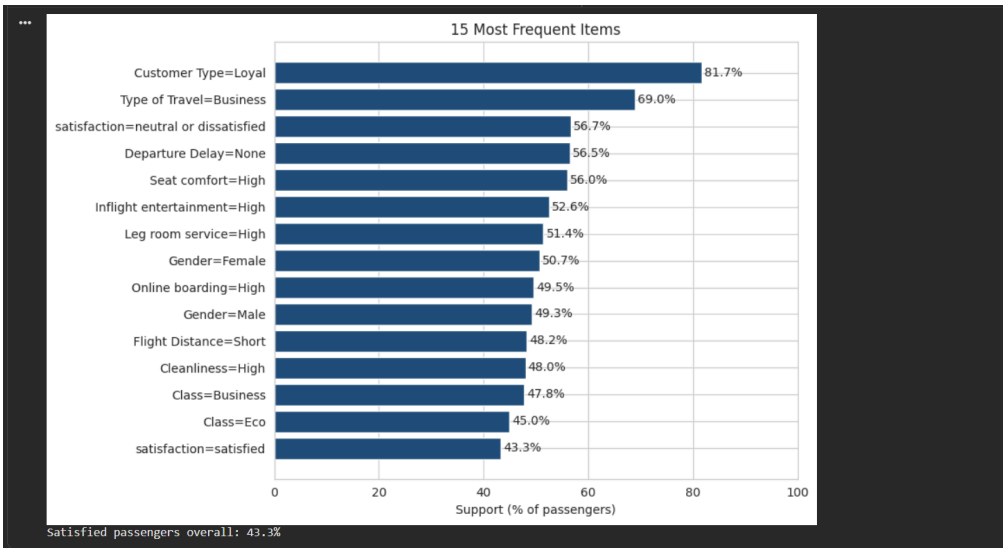


Figure 3: The 15 most frequent items

Loyal customers (81.7%) and business travellers (69.0%) are the most common items. Overall 43.3% of the passengers are satisfied. This is the base rate for judging lift: a rule that leads to satisfaction has a lift above 1 when satisfaction is more likely than this 43.3% average.

4.4 Apriori: Frequent Itemsets

```
frequent_itemsets = apriori(basket, min_support=0.10, use_colnames=True, max_len=3)
```

```
min_support = 0.1 | max itemset size = 3 | time = 1.3 s
Frequent itemsets found: 1291
itemset size         1    2    3
number of itemsets  37  357  897

Top 10 itemsets with two or more items:
```

	itemset	support
0	Customer Type=Loyal AND Type of Travel=Business	0.509
1	Customer Type=Loyal AND Seat comfort=High	0.488
2	Customer Type=Loyal AND Departure Delay=None	0.462
3	Class=Business AND Type of Travel=Business	0.457
4	Customer Type=Loyal AND Inflight entertainment=High	0.451
5	Inflight entertainment=High AND Seat comfort=High	0.444
6	Customer Type=Loyal AND Online boarding=High	0.441
7	Customer Type=Loyal AND Leg room service=High	0.433
8	Customer Type=Loyal AND satisfaction=neutral or dissatisfied	0.427
9	Cleanliness=High AND Seat comfort=High	0.420

Figure 4: Frequent itemsets by size and the ten most frequent itemsets with two or more items

With a minimum support of 0.10 and itemsets of up to 3 items, Apriori found 1,291 frequent itemsets (37 single items, 357 pairs and 897 triples) in 1.3 seconds. The most frequent pairs mostly include loyal customers, for example Loyal and Business travel (50.9%) and Loyal and High seat comfort (48.8%). Business class and business travel occur together in 45.7% of the passengers.

4.5 Association Rules

```
rules = association_rules(frequent_itemsets, metric='confidence', min_threshold=0.60)
rules.sort_values(['lift', 'confidence'], ascending=False)
```

```
Rules found (confidence >= 0.6): 1,498
Rules with lift > 1: 1,322

Top 10 rules by lift:
```

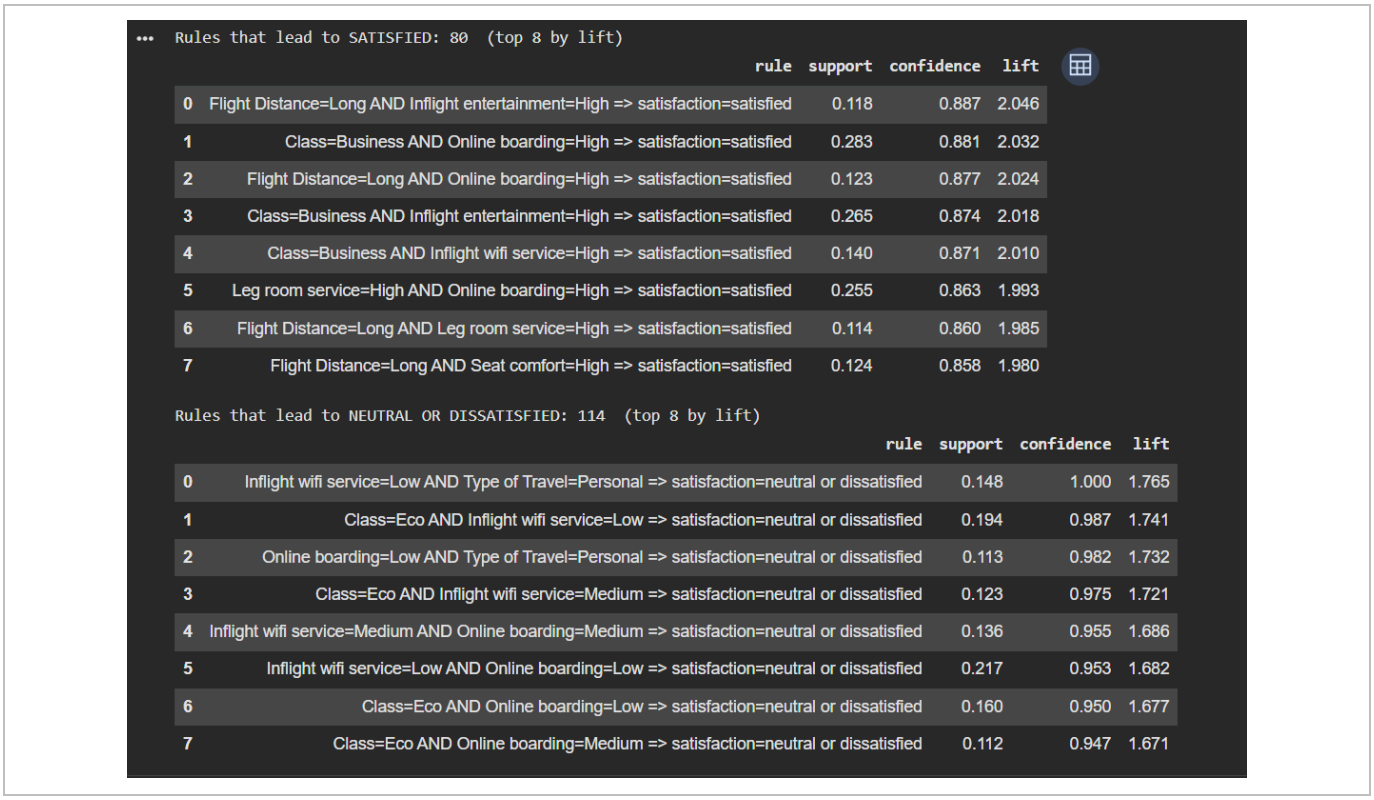
	rule	support	confidence	lift
0	Cleanliness=Medium AND Seat comfort=Medium => Inflight entertainment=Medium	0.100	0.861	4.672
1	Cleanliness=Medium AND Inflight entertainment=Medium => Seat comfort=Medium	0.100	0.761	4.231
2	Cleanliness=Medium AND satisfaction=neutral or dissatisfied => Inflight entertainment=Medium	0.102	0.762	4.135
3	Inflight entertainment=Medium AND Seat comfort=Medium => Cleanliness=Medium	0.100	0.900	3.805
4	Inflight wifi service=Medium AND satisfaction=neutral or dissatisfied => Online boarding=Medium	0.136	0.727	3.465
5	Online boarding=Medium => Inflight wifi service=Medium AND satisfaction=neutral or dissatisfied	0.136	0.646	3.465
6	Inflight entertainment=Low AND Seat comfort=Low => Cleanliness=Low	0.186	0.970	3.424
7	Cleanliness=Low => Inflight entertainment=Low AND Seat comfort=Low	0.186	0.655	3.424
8	Inflight entertainment=Low AND Online boarding=Low => Seat comfort=Low	0.107	0.876	3.374
9	Seat comfort=Medium => Inflight entertainment=Medium	0.112	0.621	3.369

Figure 5: Number of rules and the ten rules with the highest lift

With a minimum confidence of 0.60 there are 1,498 rules, and 1,322 of them have a lift above 1. The ten rules with the highest lift (3.37 to 4.67) all link service ratings of the same level, that is Medium with Medium or Low with Low. For example, passengers who rate Cleanliness and Seat comfort as Medium also rate Inflight entertainment as Medium in 86.1% of cases (lift 4.67). Some of the ten rules are the reverse of each other and have the same lift.

4.6 Rules for Satisfaction

```
sat_rules = rules[rules['consequents'] == frozenset({'satisfaction=satisfied'})]
target = frozenset({'satisfaction=neutral or dissatisfied'})
dis_rules = rules[rules['consequents'] == target]
```



Rules that lead to SATISFIED: 80 (top 8 by lift)

	rule	support	confidence	lift
0	Flight Distance=Long AND Inflight entertainment=High => satisfaction=satisfied	0.118	0.887	2.046
1	Class=Business AND Online boarding=High => satisfaction=satisfied	0.283	0.881	2.032
2	Flight Distance=Long AND Online boarding=High => satisfaction=satisfied	0.123	0.877	2.024
3	Class=Business AND Inflight entertainment=High => satisfaction=satisfied	0.265	0.874	2.018
4	Class=Business AND Inflight wifi service=High => satisfaction=satisfied	0.140	0.871	2.010
5	Leg room service=High AND Online boarding=High => satisfaction=satisfied	0.255	0.863	1.993
6	Flight Distance=Long AND Leg room service=High => satisfaction=satisfied	0.114	0.860	1.985
7	Flight Distance=Long AND Seat comfort=High => satisfaction=satisfied	0.124	0.858	1.980

Rules that lead to NEUTRAL OR DISSATISFIED: 114 (top 8 by lift)

	rule	support	confidence	lift
0	Inflight wifi service=Low AND Type of Travel=Personal => satisfaction=neutral or dissatisfied	0.148	1.000	1.765
1	Class=Eco AND Inflight wifi service=Low => satisfaction=neutral or dissatisfied	0.194	0.987	1.741
2	Online boarding=Low AND Type of Travel=Personal => satisfaction=neutral or dissatisfied	0.113	0.982	1.732
3	Class=Eco AND Inflight wifi service=Medium => satisfaction=neutral or dissatisfied	0.123	0.975	1.721
4	Inflight wifi service=Medium AND Online boarding=Medium => satisfaction=neutral or dissatisfied	0.136	0.955	1.686
5	Inflight wifi service=Low AND Online boarding=Low => satisfaction=neutral or dissatisfied	0.217	0.953	1.682
6	Class=Eco AND Online boarding=Low => satisfaction=neutral or dissatisfied	0.160	0.950	1.677
7	Class=Eco AND Online boarding=Medium => satisfaction=neutral or dissatisfied	0.112	0.947	1.671

Figure 6: The strongest rules that lead to satisfied and to neutral or dissatisfied passengers

Of the 1,498 rules, 80 lead to 'satisfied' and 114 lead to 'neutral or dissatisfied'. The strongest rules for satisfied passengers combine a long flight or Business class with a High rating for online boarding, inflight entertainment or wifi. For example, Business class with High online boarding is satisfied in 88.1% of cases (lift 2.03) and applies to 28.3% of all passengers. The strongest rules for neutral or dissatisfied passengers combine a Low or Medium wifi or online boarding rating with Eco class or personal travel, with confidence between 94.7% and 100%. A lift of 1.76 is the highest possible for this result, because 1 / 0.567 = 1.76.

4.7 Support against Confidence

```
plt.scatter(rules['support'], rules['confidence'], c=rules['lift'], cmap='viridis')
```

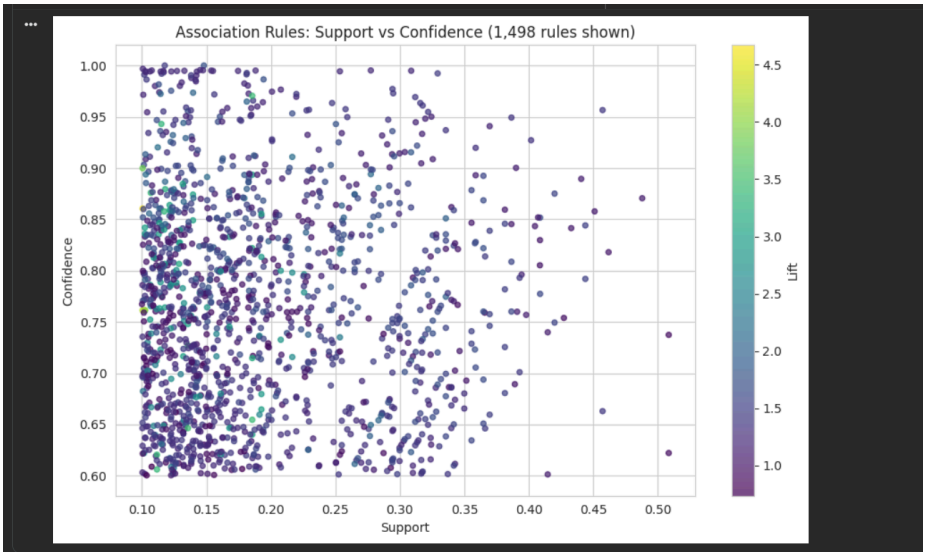


Figure 7: All 1,498 rules plotted by support and confidence, coloured by lift

Most rules have a low support between 0.10 and 0.20. The rules with the highest lift (green and yellow dots) are all among these low-support rules, while the rules with the highest support (above 0.4) have a low lift. So the strongest associations apply to a smaller share of the passengers.

4.8 Effect of the Minimum Support



Figure 8: Number of itemsets, number of rules and time for different minimum support values

Lowering the minimum support from 0.30 to 0.10 increases the number of frequent itemsets from 101 to 1,291 (about 13 times) and the number of rules from 193 to 1,498 (about 8 times), and the time from 0.1 to 1.3 seconds. A low minimum support finds more rules, including less common ones, but also takes longer and produces many rules to examine.

4.9 Key Findings

```
Overall, 43.3% of passengers are satisfied.

Strongest rules for SATISFIED passengers:
 * IF Flight Distance=Long AND Inflight entertainment=High  THEN satisfaction=satisfied
     support 11.8% | confidence 88.7% | lift 2.05
 * IF Class=Business AND Online boarding=High  THEN satisfaction=satisfied
     support 28.3% | confidence 88.1% | lift 2.03
 * IF Flight Distance=Long AND Online boarding=High  THEN satisfaction=satisfied
     support 12.3% | confidence 87.7% | lift 2.02
 * IF Class=Business AND Inflight entertainment=High  THEN satisfaction=satisfied
     support 26.5% | confidence 87.4% | lift 2.02

Strongest rules for NEUTRAL OR DISSATISFIED passengers:
 * IF Inflight wifi service=Low AND Type of Travel=Personal  THEN satisfaction=neutral or dissatisfied
     support 14.8% | confidence 100.0% | lift 1.76
 * IF Class=Eco AND Inflight wifi service=Low  THEN satisfaction=neutral or dissatisfied
     support 19.4% | confidence 98.7% | lift 1.74
 * IF Online boarding=Low AND Type of Travel=Personal  THEN satisfaction=neutral or dissatisfied
     support 11.3% | confidence 98.2% | lift 1.73
 * IF Class=Eco AND Inflight wifi service=Medium  THEN satisfaction=neutral or dissatisfied
     support 12.3% | confidence 97.5% | lift 1.72
```

Figure 9: The strongest rules written in plain English

5. Result Summary

Item	Result
Transactions	103,904 passengers
Items	39 items from 14 columns (13.9 items per transaction on average)
Parameters	min_support = 0.10, max itemset size = 3, min_confidence = 0.60
Frequent itemsets	1,291 (37 single items, 357 pairs, 897 triples)
Association rules	1,498, of which 1,322 have lift above 1
Rules leading to satisfied	80
Rules leading to neutral or dissatisfied	114
Time at min_support 0.10	1.3 seconds

Rule	Support	Confidence	Lift
IF Class=Business AND Online boarding=High THEN satisfied	28.3%	88.1%	2.03
IF Class=Business AND Inflight entertainment=High THEN satisfied	26.5%	87.4%	2.02
IF Flight Distance=Long AND Inflight entertainment=High THEN satisfied	11.8%	88.7%	2.05
IF Inflight wifi service=Low AND Type of Travel=Personal THEN neutral or dissatisfied	14.8%	100.0%	1.76
IF Class=Eco AND Inflight wifi service=Low THEN neutral or dissatisfied	19.4%	98.7%	1.74
IF Inflight wifi service=Low AND Online boarding=Low THEN neutral or dissatisfied	21.7%	95.3%	1.68

6. Conclusion

The Apriori algorithm was implemented in Python on 103,904 airline passenger transactions. With a minimum support of 0.10 and a minimum confidence of 0.60 it found 1,291 frequent itemsets and 1,498 association rules. Satisfaction is most strongly associated with Business class or long flights combined with High online boarding or inflight entertainment ratings (confidence about 88%, lift about 2.0), and dissatisfaction with Low or Medium wifi or online boarding ratings together with Eco class or personal travel (confidence 95% to 100%). A lower minimum support finds many more rules (193 at 0.30 and 1,498 at 0.10) but takes longer. These findings agree with the important features found in Experiment 4.